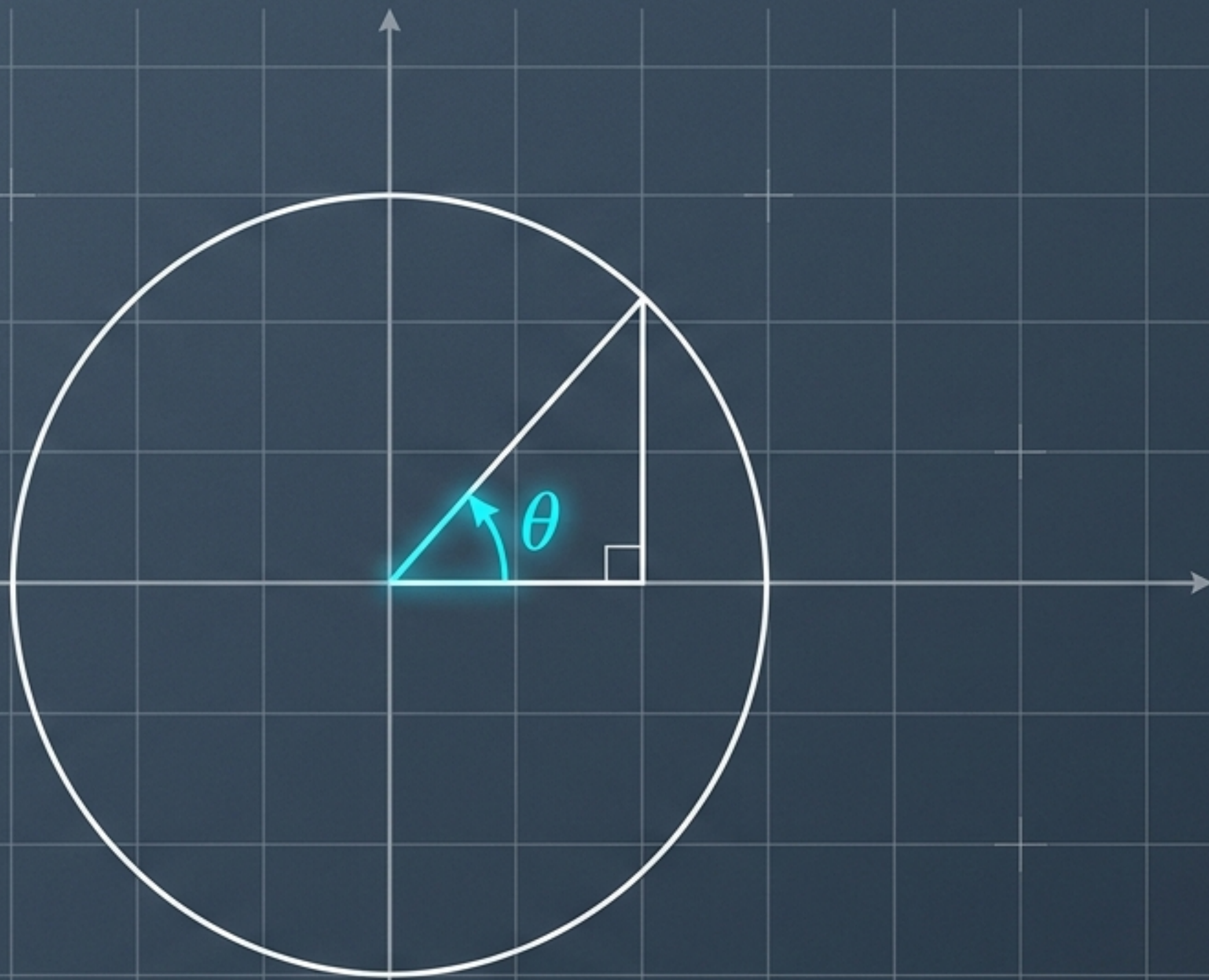


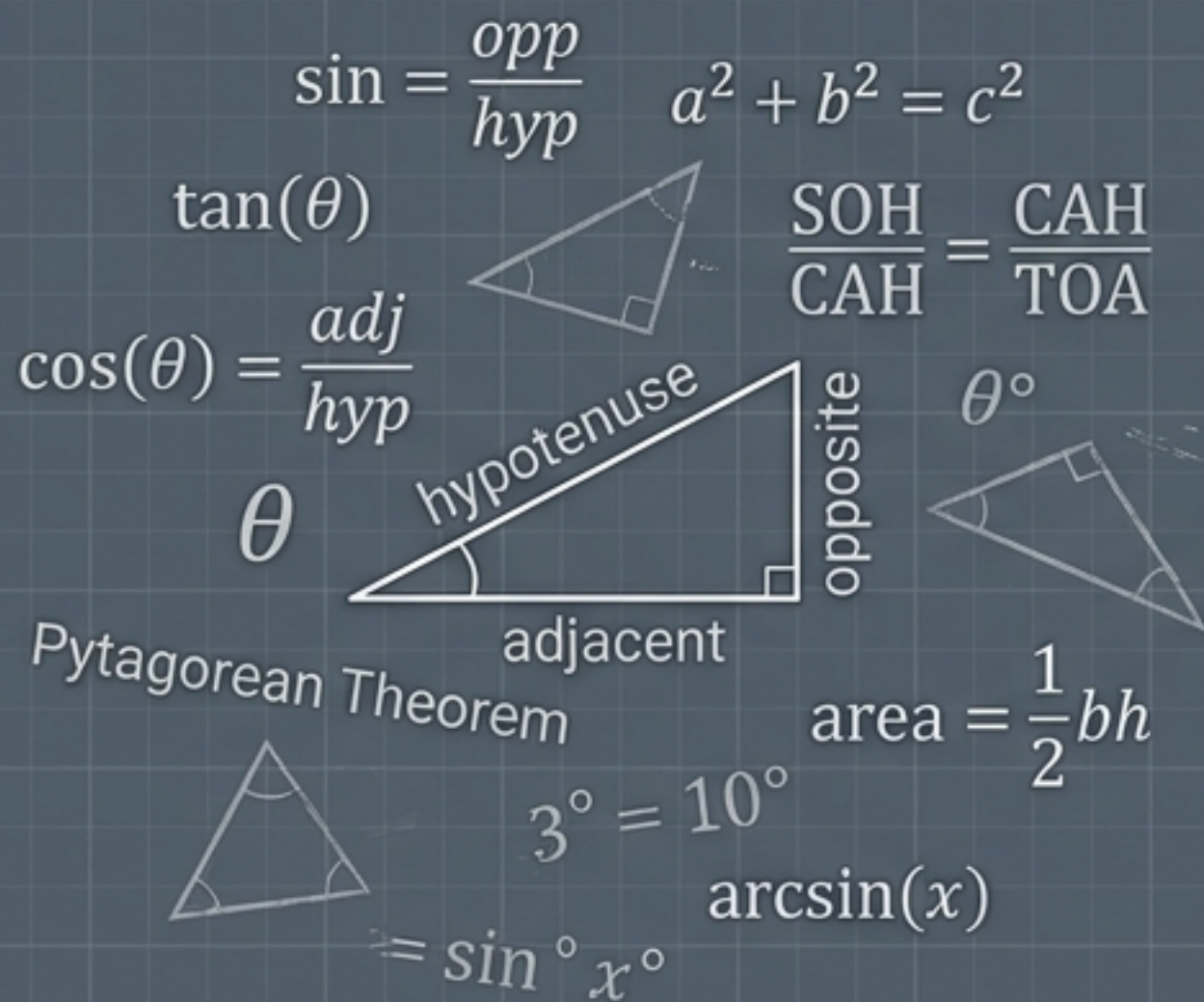


Trigonometry and Transformation

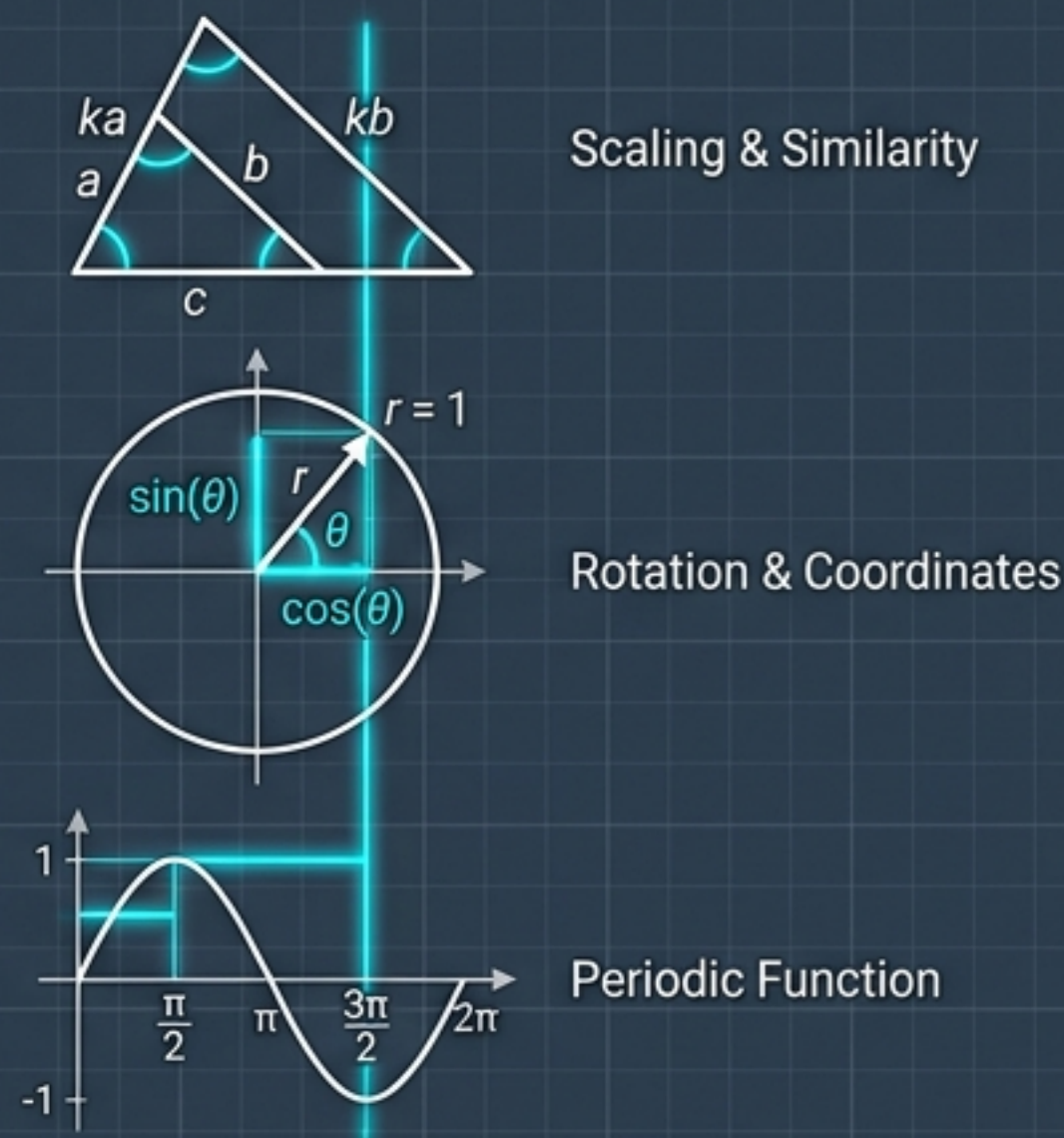
The Geometry of Invariance

A Search for Structure

Trigonometry is an extended investigation into the interaction between transformation and invariance. It asks a deceptively simple question: what remains unchanged when everything else changes?



The Procedural View: Isolated Formulas



The Structural View: Transformation & Invariance

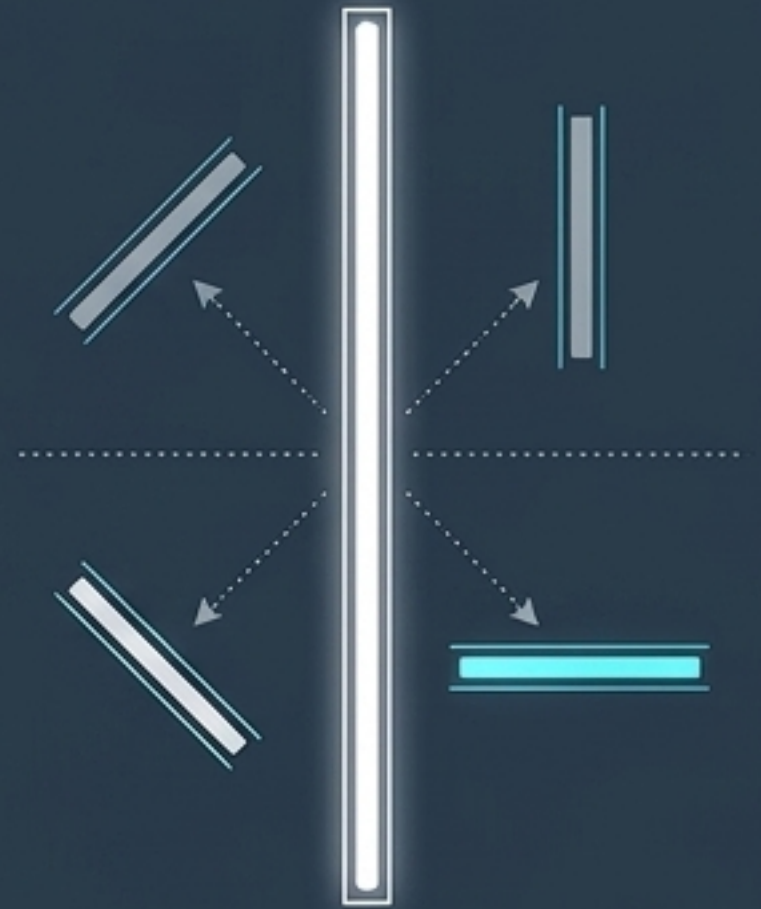
Takeaway: We do not study triangles just to measure them. We study them to isolate the **invariants**—the hidden rules that survive a universe in motion.

Constraint Generates Structure

A space of possibilities remains formless until rules are applied. It is the constraint, not the alphabet, that produces structure worth studying.



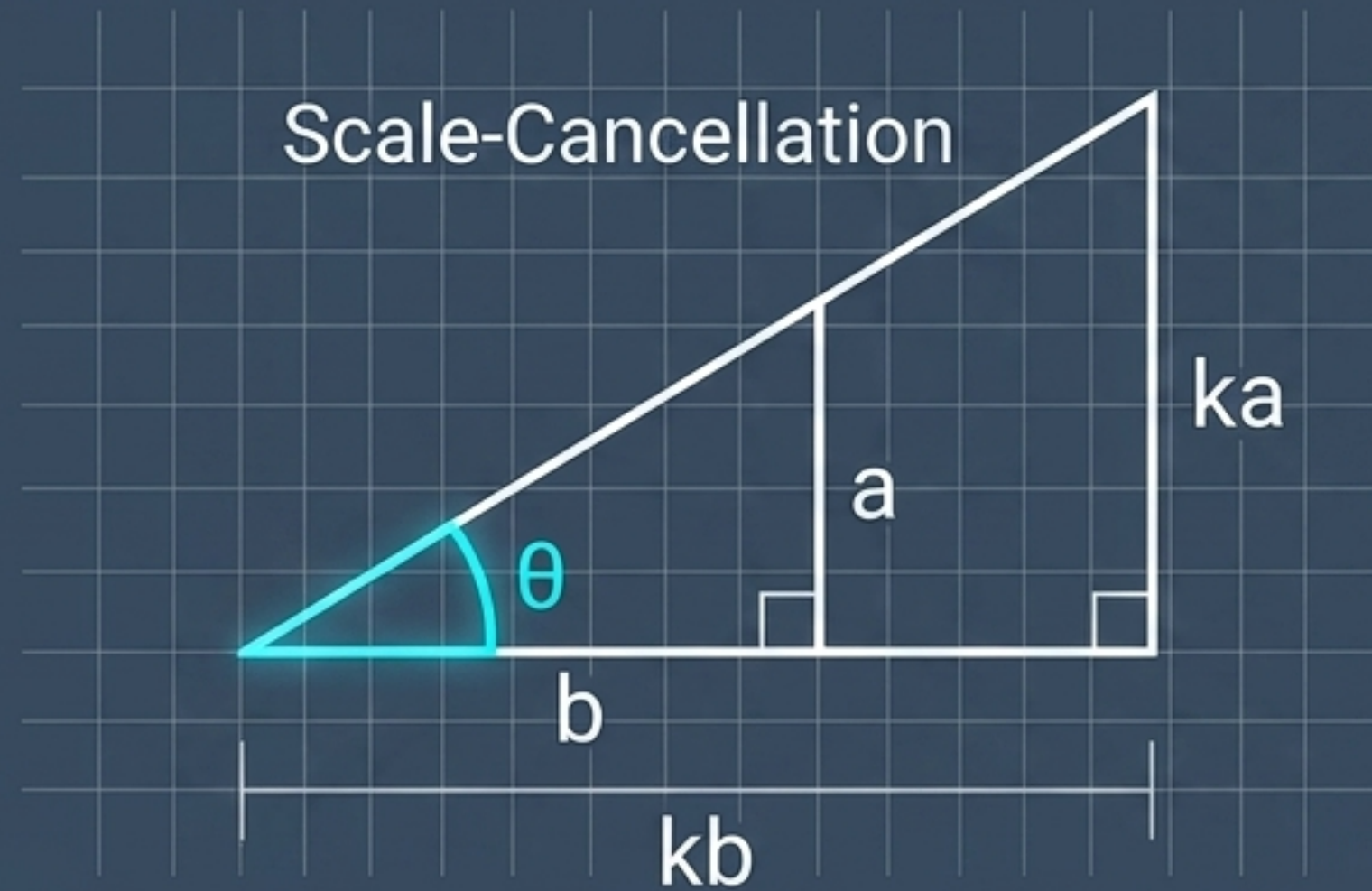
Cistercian Numerals & Composition



Medieval monks generated 10,000 representable values by constraining strokes to 4 independent positions. The power comes not from the symbols, but from how independent coordinates combine—a principle that mirrors the very vectors we use to map space.

The Geometry of Scale

A shape may grow or shrink, but its internal relationships persist. Whenever a figure changes size uniformly, the scale factor cancels itself out. The scale factor vanishes, and what remains is pure structure.

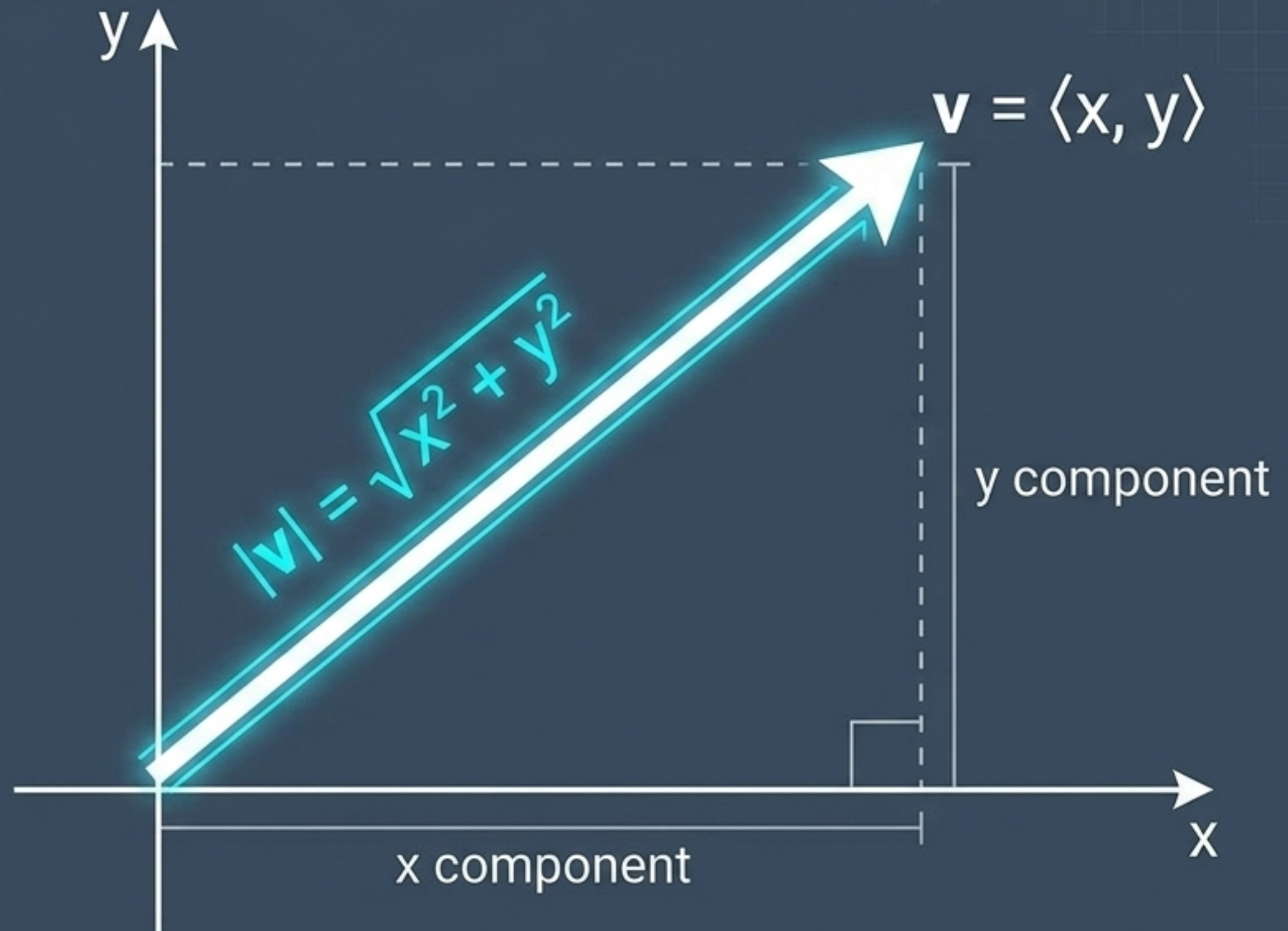


$$\frac{\cancel{k}a}{\cancel{k}b} \rightarrow \frac{a}{b}$$

Scale Factor Cancels

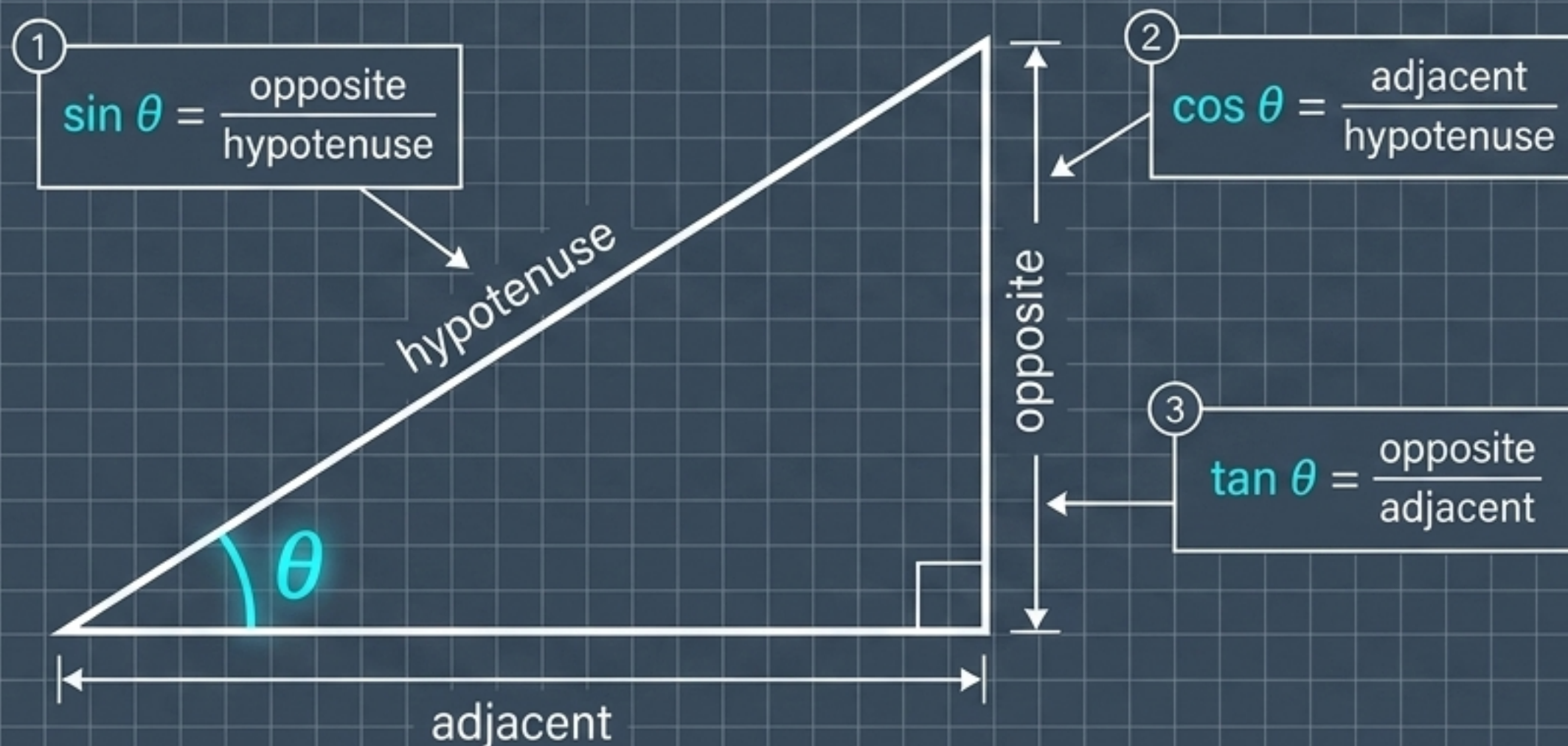
Displacement and the Hidden Triangle

A point describes a location, but a vector describes movement. To capture both magnitude and direction, every vector inherently generates a right triangle. This right triangle is the bridge between length and direction.

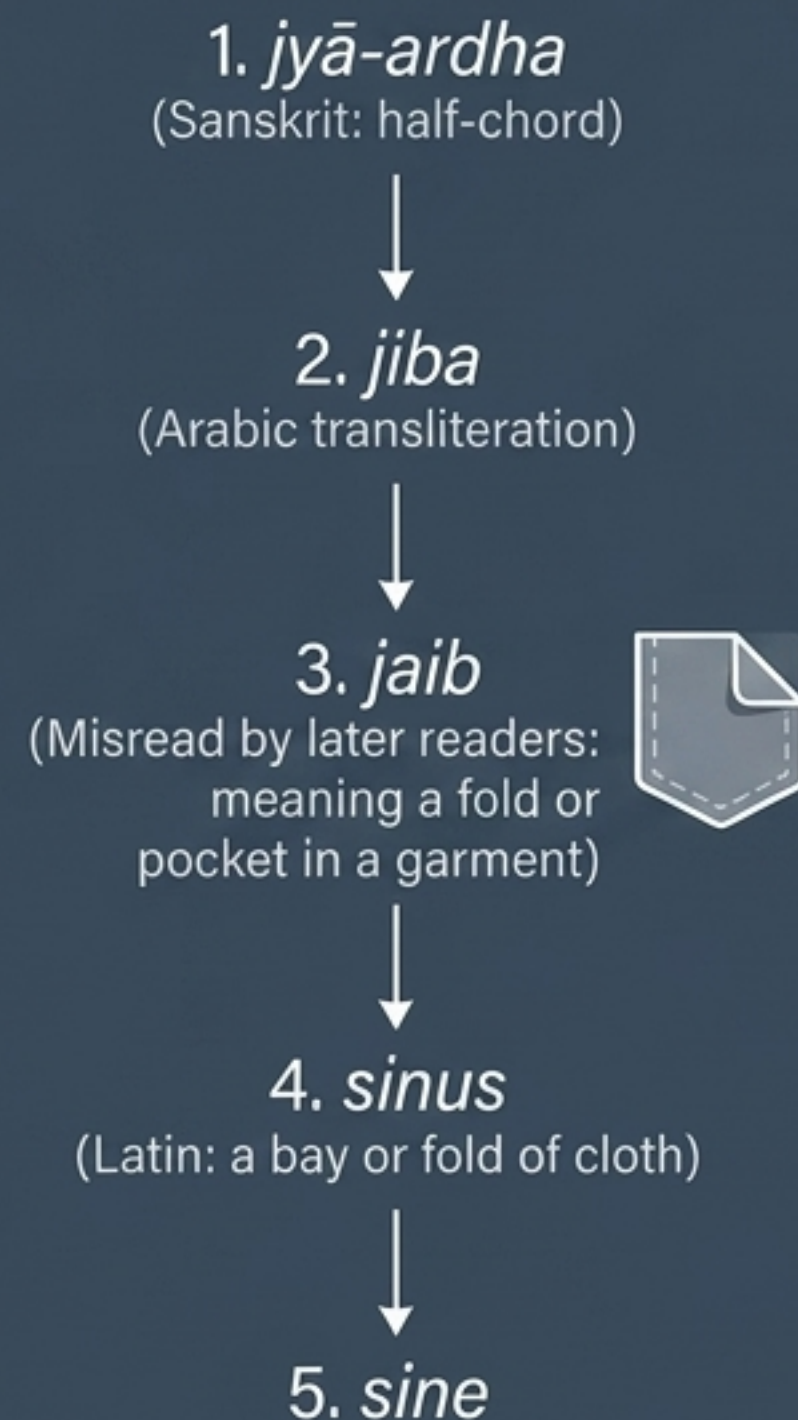


Naming the Invariants

We name quantities when they survive irrelevant changes. Sine, cosine, and tangent are not arbitrary formulas; they are simply the names we give to the ratios that survive scaling.

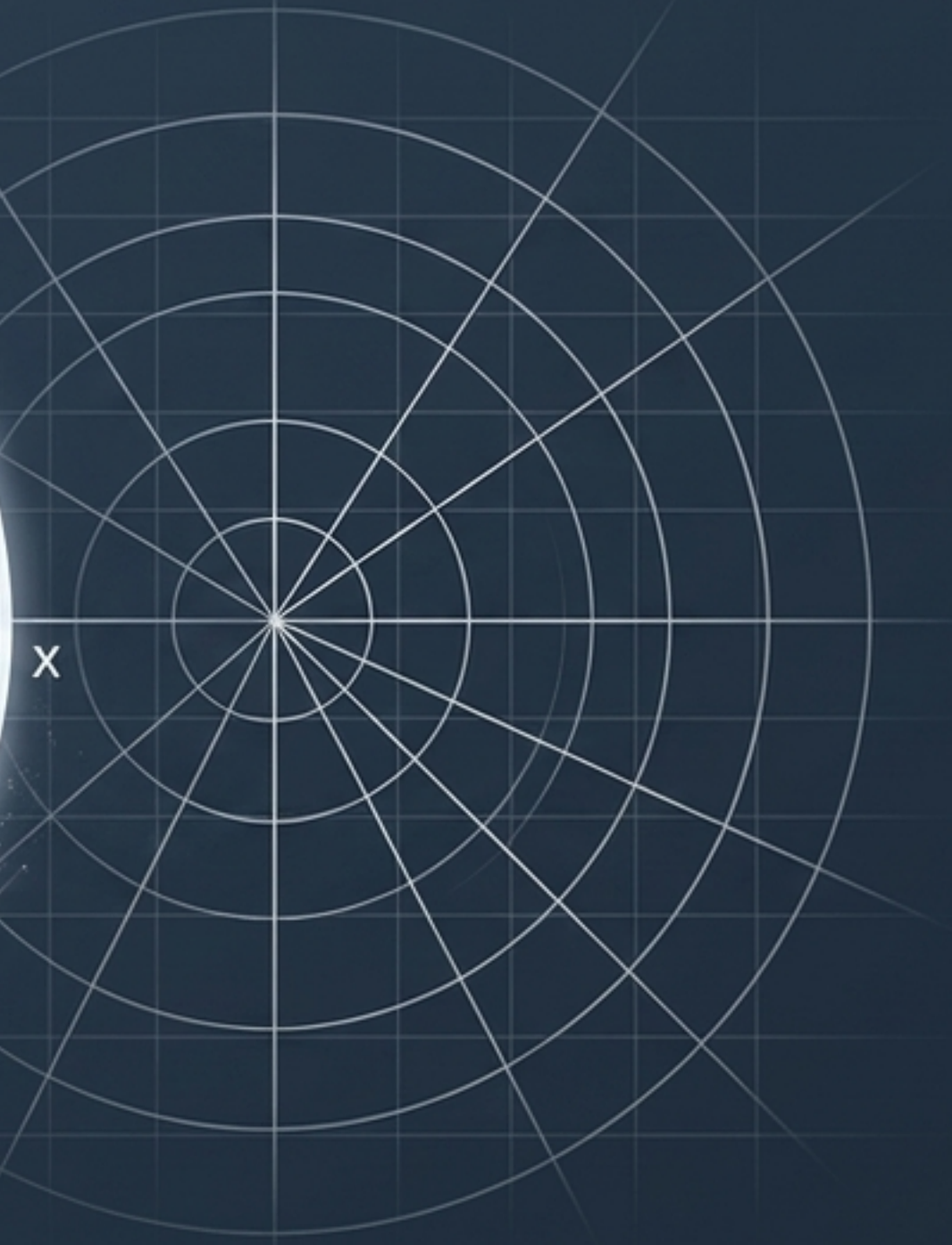
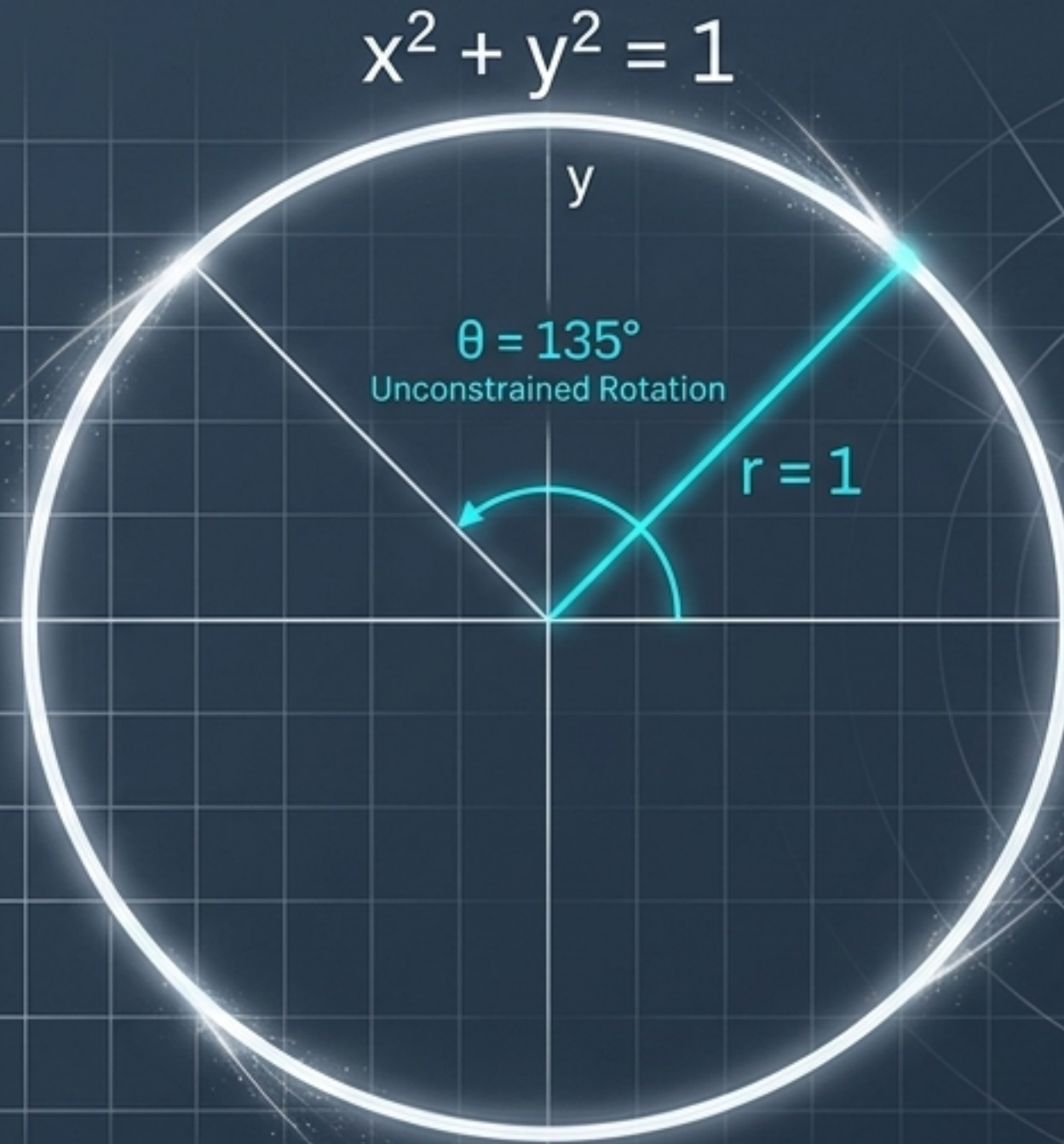


A Translator's Misreading



Breaking the Triangle Constraint

A right triangle can only hold acute angles ($0^\circ < \theta < 90^\circ$). To describe universal direction, we must remove the triangle entirely. The unit circle isolates directional information by fixing magnitude at 1, allowing rotation to be completely unconstrained.



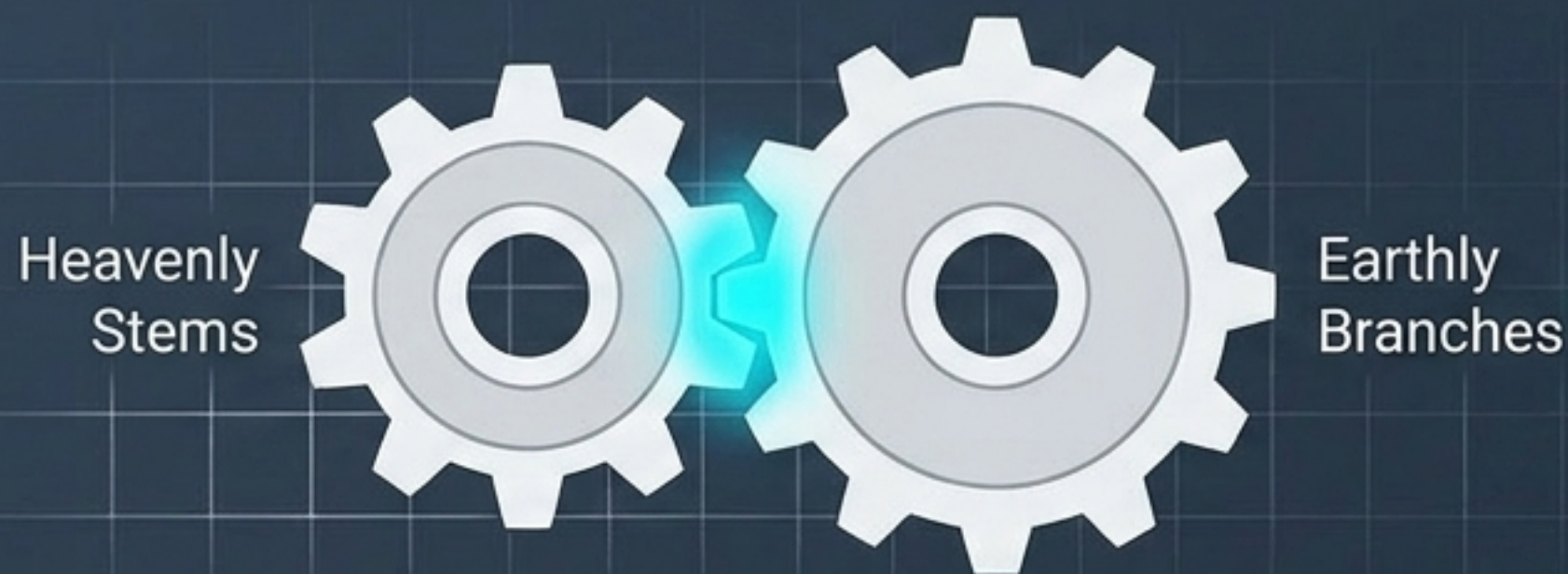
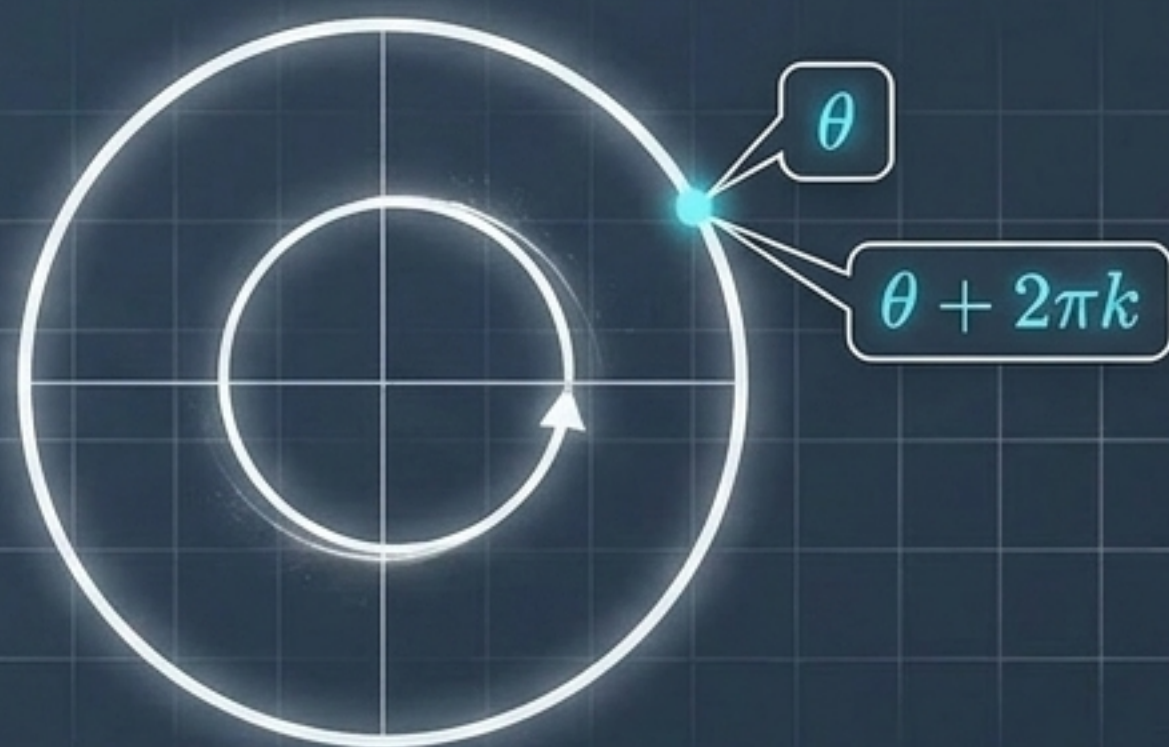
Geometry of Measurement vs. Geometry of Structure

	Degrees	Radians
Underlying Logic	Based on human measurement and arbitrary history (Babylonian astronomy).	Based on mathematical structure. A radian perfectly matches the arc length to the radius.
Dimensions	360° for a full circle.	2π for a full circle.
Arc Length Equation	Messy and procedural: $s = \frac{\theta \cdot \pi \cdot r}{180}$	Structurally pure: $s = r\theta$

Radians are not just another unit; they are the **native language of circles**.

Periodicity and Coterminal Cycles

Continuous rotation means the geometry is inherently cyclic. Adding a full revolution (2π) changes the description, but not the physical reality.



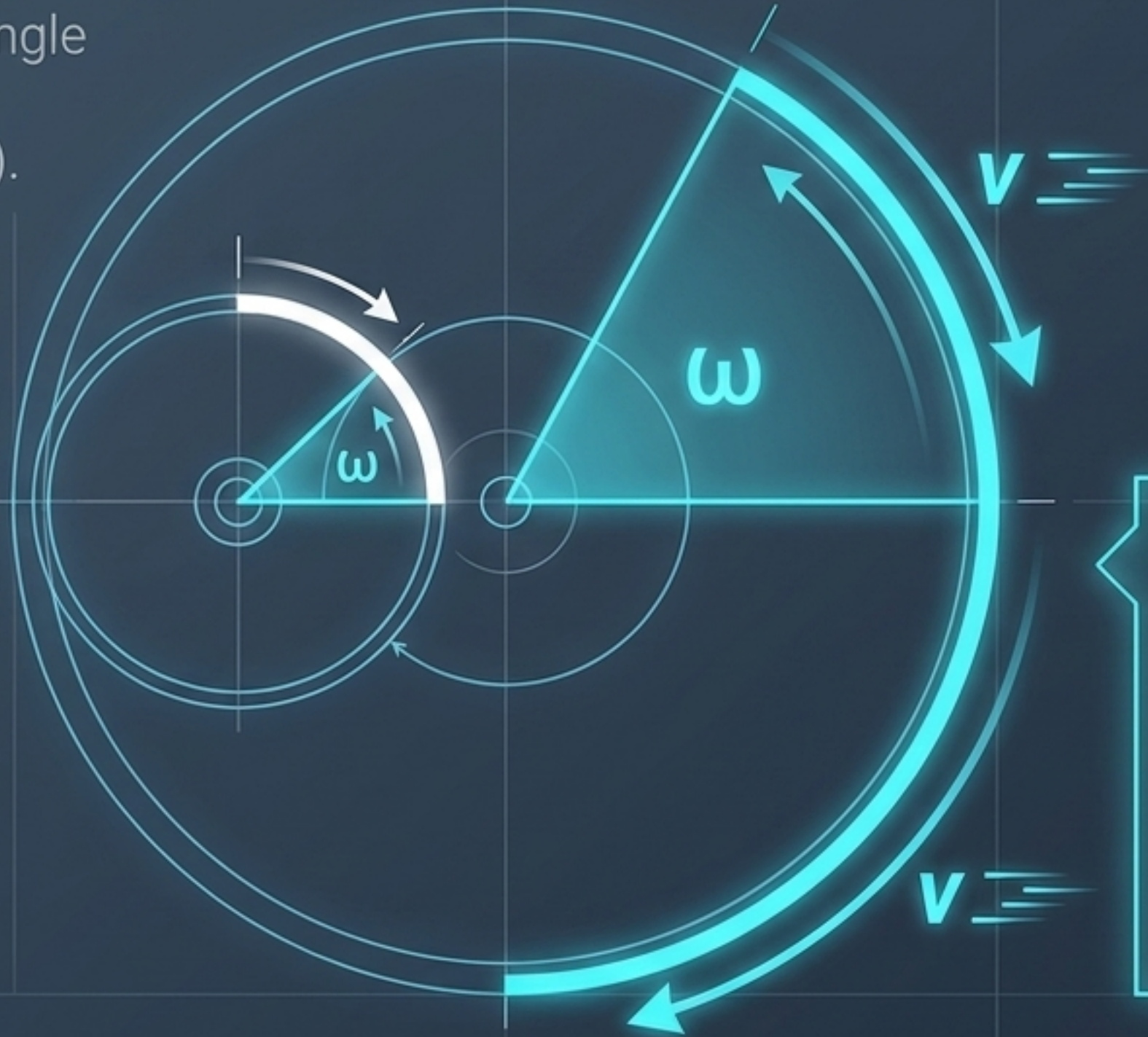
Simultaneous Periods: Centuries before modular arithmetic, the Chinese Sexagenary cycle tracked time by pairing a **10-symbol cycle** with a **12-symbol cycle**. They return to their start simultaneously every **60 steps**—the exact same **cyclic interaction** found in harmonic math.

Two Eras of Trigonometry

	Right Triangles	Unit Circle
Definition of Angle	Angle is constrained inside a geometric shape ($0 < \theta < 90$).	Angle is an unconstrained, continuous rotation ($-\infty < \theta < \infty$).
Output Bounds	Outputs are strictly positive side lengths.	Outputs are cyclical coordinates on a plane (positive or negative).
Core Function	Used to calculate static geometric distances and heights.	Used to model continuous, cyclical state spaces and waves.

When Angles Move

Once time is introduced, an angle is no longer a fixed quantity, but a continuous function $\theta(t)$.



Angular velocity ($\omega = \theta/t$) measures the rate of rotation. But linear velocity requires the radius multiplier ($v = r\omega$).

A larger radius produces a faster linear speed, even when the rotational speed is identical.

The Unification of Motion and Geometry

An oscillation—a swinging pendulum, a sound wave, a vibrating string—is ultimately just the shadow of a circle turning at constant speed.

Invariant Ratios (Ch 3)

Unit Circle Starting Angles (Ch 4)

$$x(t) = \cos(\omega t + \theta_0)$$

Coordinates (Ch 2)

Continuous Motion & Time (Ch 5)

Projection of Circular Motion

